

Lecture 5

Review Questions

Logits and softmax

A classifier produces logits $\mathbf{z} = [2.0, 1.0, -1.0]$.

Suppose the same constant 100 is added to every logit. What changes?

- A. The predicted class changes.
- B. The softmax probabilities change, but the predicted class does not.
- C. Neither the softmax probabilities nor the predicted class changes.
- D. The logits become probabilities.

Answer: C

Softmax depends on differences between logits:

$$\text{softmax}(z_k + c) = \text{softmax}(z_k).$$

This also illustrates why logits are relative class scores rather than absolute probabilities.

$$p_k = \frac{e^{z_k}}{\sum_j e^{z_j}}.$$

$$\begin{aligned} p'_k &= \frac{e^{z_k+c}}{\sum_j e^{z_j+c}} \\ &= \frac{e^c e^{z_k}}{e^c \sum_j e^{z_j}} \\ &= \frac{e^{z_k}}{\sum_j e^{z_j}} = p_k. \end{aligned}$$

Equivariance versus invariance

An object is translated to a different location in an image. What behaviour would we ideally expect from a CNN classifier?

- A. Every intermediate feature map should remain identical.
- B. Intermediate feature maps should translate, while the final class prediction remains approximately unchanged.
- C. Intermediate feature maps should remain unchanged, while the class prediction translates.
- D. Both the feature maps and class prediction must change.

Answer: B

Convolution produces approximately translation-**equivariant** feature maps: shifting the input shifts the responses. Spatial aggregation and the classification head can then produce an approximately translation-**invariant** class prediction.

Two different IoU thresholds

Object detection commonly uses an IoU threshold during both NMS and evaluation. What is the important difference?

- A. NMS compares predictions with ground truth; evaluation compares predictions with each other.
- B. Both thresholds compare predictions with ground truth.
- C. NMS compares predictions with each other; evaluation compares predictions with ground truth.
- D. Both thresholds perform exactly the same operation.

Answer: C

NMS uses overlap between predicted boxes to remove likely duplicates. Evaluation uses overlap between a prediction and a ground-truth box to decide whether the prediction is sufficiently well localized.

Evolution of deep object detectors

Which description is correct?

- A. R-CNN processes the image once, while Fast R-CNN processes every proposal separately.
- B. Fast R-CNN shares convolutional computation but still depends on externally generated proposals; Faster R-CNN learns the proposals with a region proposal network.
- C. Faster R-CNN is a one-stage detector because its proposals are learned.
- D. YOLO first generates region proposals and then classifies each proposal separately.

Answer: B

The progression is:

- **R-CNN:** run a CNN separately on each proposed region.
- **Fast R-CNN:** compute one shared image feature map, but retain external proposals.
- **Faster R-CNN:** generate proposals inside the network with an RPN.
- **YOLO:** make dense class and box predictions in a one-stage architecture.